



Smallest frameless BLDC motor

The 38BLF07 is the Delta Line's most miniature frameless BLDC motor in its series, with a 38mm outer diameter. The motor, measuring only 14mm in length, offers a rated torque of 0.1Nm at a speed



of 8000rpm and can reach a peak torque of 0.3Nm. Weighing just 57gm, it is designed to have minimal inertia, making it well-suited for applica-

tions requiring rapid acceleration and deceleration. It is designed to address the spatial and performance requirements of handheld power tools, precision medical instruments, and space-restricted robotics, where the key factors are torque, efficiency, and a compact form factor. The frameless design allows seamless integration directly into devices or machinery, ensuring maximum efficiency while minimising the space footprint.

Delta Line

<https://www.delta-line.com>

World's smallest and thinnest multilayer ceramic capacitors

Murata has launched the LLC series of automotive-grade ceramic capacitors. These components feature reversed termination for low ESL and come in the



world's first 0.18mm profile package. The footprint is only $0.5 \times 1.0\text{mm}$ for $1\mu\text{F}$ capacitance, making them the smallest in the market. The LLC series uses homogenisation techniques to achieve about 20% reduction in component height compared with existing parts. This enables easier placement of capacitors on the backside of circuit boards, even among solder ball terminations, optimising their position for processor power

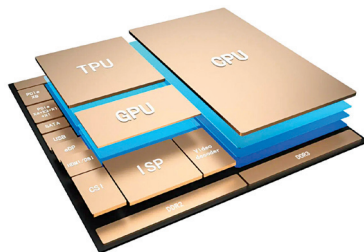
rail decoupling, reducing the need for capacitors, lowering costs, and improving system reliability. The capacitor meets the requirements of modern, low-voltage, compute-intensive applications such as automotive advanced driver-assistance systems (ADAS).

Murata

<https://www.murata.com>

Customisable RISC-V cores for AI computing processors

Sophgo has licensed the SiFive RISC-V high-performance processor cores, including the SiFive Performance P670 and SiFive Intelligence X280, to develop RISC-V AI computing processors. The SG2380 is a 2.5 GHz 16-core



RISC-V processor that utilises the SiFive Performance P670 with a multi-cluster, multi-core configuration. It offers high performance and advantages in RISC-V vector computation. Additionally, the SG2380 incorporates the SiFive Intelligence X280 and a Vector Cooperative Processor Interface Extension (VCIX), integrating Sophgo's AI/ML hardware design. With comprehensive software toolchain support from RISC-V, this coprocessor is designed to cater to not only AI/ML but various advanced computing requirements.

Sophgo & SiFive

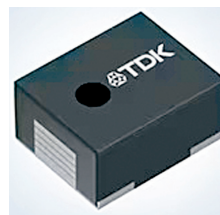
<https://www.sophon.ai>

<https://www.sifive.com>

Industry's lowest profile inductors for power circuits

TDK Corporation has launched the PLEA85 series of high-efficiency power inductors designed for battery-powered wearables and other devices to enhance

operational durations. This series boasts the industry's slimmest profile, achieved through TDK's novel low-loss magnetic material and thin-film process-



ing techniques. Main applications include wearable devices like true wireless stereo (TWS) earphones, hearing aids, smartwatches,

and small-sized power supply modules. Measuring 1.0mm (L) $\times$ 0.8mm (W) $\times$ 0.55mm (H), the PLEA85 series facilitates design miniaturisation, optimising the utilisation of low-profile ICs. The bottom electrode and partly L-shaped form on the side enhance high-density surface mounting, minimising assembly errors and bolstering terminal durability for a more resilient end product.

TDK Corporation

<https://www.tdk.com>

IoT connectivity SoC with Wi-Fi 6E and Bluetooth 5.3

Synaptics has introduced the SYN43711, expanding its wireless connectivity offerings for the Internet of Things (IoT). This system-on-chip (SoC) combines Wi-Fi 6E and Bluetooth



5.3, enhancing its product lineup for the IoT market segment. The SoC is designed for use cases requiring

Wi-Fi throughput, range, and network compatibility. It is intended for applications like home appliances, surveillance cameras, robots, and various enterprise systems. Notable features include Wi-Fi sensing support and utilising a proprietary algorithm to analyse channel state information. This algorithm enables any Wi-Fi-connected embedded device to detect human presence, estimate motion, approximate location, and sense subtle gestures and movements, such as breathing.

Synaptics

<https://www.synaptics.com>